

# Forecasting with the Help of Survey Information

**Federica Brenna, Žymantas Budrys**

Lietuvos bankas and Vilnius University

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BEC, Riga

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- Professional forecasters benefit from:
  - Access to extensive data
  - Timely information updates
  - Use of informal methods (e.g., judgement, qualitative information)
- **Econometric models** still essential: provide internal consistency, produce density forecasts.
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**This paper:** Can we efficiently incorporate surveys into forecasting models to improve *real time* forecasts?

Ex-post adjustment: estimate a model, produce forecasts with it, *then* adjust them with external information. Examples include:

- *Entropic tilting*: Galvão et al. (2021), Ganics and Odendahl (2021), Robertson et al. (2005), and Tallman and Zaman (2020)
- *Optimal pooling*: Ang et al. (2007) and Bańbura et al. (2021)
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**This paper's idea:** incorporate survey forecasts *directly into* a reduced form model in order to get more precise estimates, and, therefore, improve forecasts.

- **Use surveys during estimation:** Survey forecasts enter a Bayesian VAR as informative signals about the economy, rather than being used only after estimation.
- **Develop a parsimonious framework:** Adding survey forecasts does not increase the number of parameters to be estimated, rather adds “observations” to the system, improving estimation’s precision.
- **Test in real-time setting:** We evaluate the approach using ECB SPF data and real-time euro area forecasts through multiple crises.
- **Improve medium-term forecasting:** Incorporating survey information yields better forecasts, particularly at 1–2 year horizons and for predictive densities.

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- **This paper**:  
Estimate VAR on data **and** surveys → Obtain new estimated parameters incorporating survey information → Produce forecasts
- The two approaches estimate the same **number** of parameters, but the second one incorporates the **additional signal** from surveys.

Starting point is a **VAR(p)+SV**:

$$y_t = c + \beta_1 y_{t-1} + \dots + \beta_p y_{t-p} + A_0^{-1} \mathbf{\Lambda}_t e_t \quad (1)$$

Starting point is a **VAR(p)+SV**:

$$y_{t-1} = c + \beta_1 y_{t-2} + \dots + \beta_p y_{t-p-1} + A_0^{-1} \Lambda_{t-1} e_{t-1} \quad (1)$$

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$$\text{horizon } h: \quad y_{t+h|t} = c + \beta_1 y_{t+h-1|t} + \dots + \beta_p y_{t+h-p|t} + A_0^{-1} \Lambda_{t+h|t} e_{t+h|t} \quad (3)$$

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Observed forecasts decomposed into **model-consistent** and **judgement** component (varying across time and horizons, encompassing all information not captured by the VAR)

Matrix form

SV

## ECB Survey of Professional Forecasters and ECB Real-Time Database

- Sample 1990q1-2024q3
- **4 variables:** real GDP growth, unemployment rate, HICP inflation, 3-m Euribor
- At each quarter  $q$ , observed data (all four variables) and forecasts (*only* for GDP and HICP) at different horizons.
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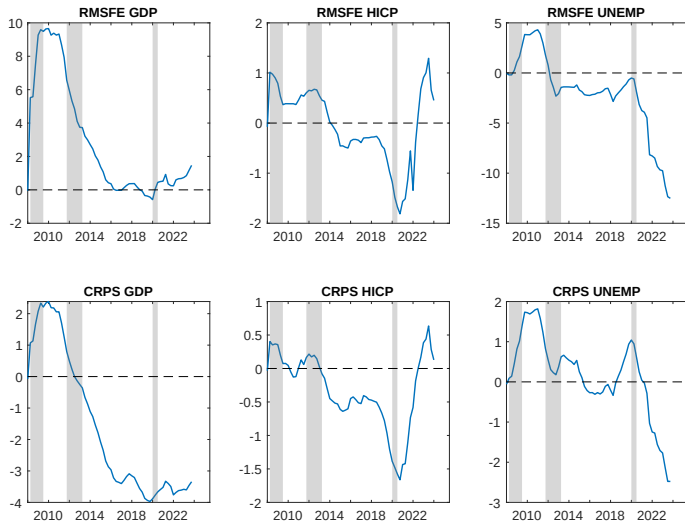
## Evaluation:

- Recursive estimation from **2007q2**; forecast evaluation from **2007q3 to 2024q3**
- Comparison of standard BVAR+SV and augmented model
- Point accuracy, density accuracy, and calibration

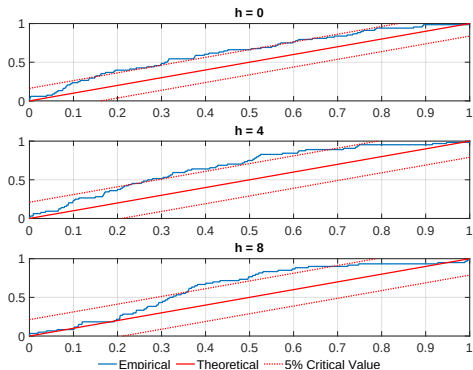
| Horizon             | h=0         | h=1         | h=2         | h=3         | h=4         | h=5         | h=6         | h=7         | h=8         |
|---------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <i>GDP</i>          |             |             |             |             |             |             |             |             |             |
| RMSFE               | 1.05        | <b>0.99</b> | 1.01        | <b>0.99</b> | 1.00        | 1.00        | <b>0.99</b> | <b>0.99</b> | <b>0.98</b> |
| CRPS                | <b>0.98</b> | <b>0.95</b> | <b>0.94</b> | <b>0.90</b> | <b>0.94</b> | <b>0.94</b> | <b>0.91</b> | <b>0.89</b> | <b>0.92</b> |
| <i>HICP</i>         |             |             |             |             |             |             |             |             |             |
| RMSFE               | 1.23        | 1.05        | 1.03        | <b>0.96</b> | 1.01        | <b>0.98</b> | <b>0.98</b> | <b>0.97</b> | <b>0.98</b> |
| CRPS                | 1.13        | 1.03        | 1.00        | <b>0.96</b> | 1.01        | <b>0.97</b> | <b>0.95</b> | <b>0.94</b> | <b>0.95</b> |
| <i>Unemployment</i> |             |             |             |             |             |             |             |             |             |
| RMSFE               | 1.07        | 1.03        | <b>0.96</b> | <b>0.91</b> | <b>0.90</b> | <b>0.90</b> | <b>0.90</b> | <b>0.90</b> | <b>0.90</b> |
| CRPS                | 1.02        | 1.04        | <b>0.98</b> | <b>0.94</b> | <b>0.93</b> | <b>0.93</b> | <b>0.93</b> | <b>0.93</b> | <b>0.93</b> |

**Note:** The table shows accuracy metrics in relative terms, with the BVAR model at the denominator and the Survey-augmented BVAR (S-BVAR) at the numerator. Values smaller than one indicate that the latter model is more accurate on average over the sample for the respective horizon.

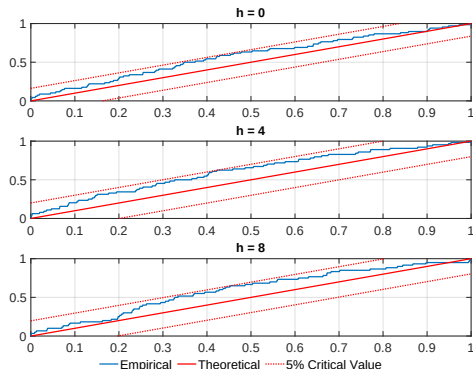
# Forecast performance over time: 1-year-ahead



(a) Baseline



(b) S-BVAR



- 1 Survey expectations contain valuable information that can be exploited during model estimation.
- 2 We propose a tractable framework that incorporates that information directly into a Bayesian VAR with stochastic volatility.
- 3 Using real-time euro area data, we find that this approach improves forecasting performance, particularly at medium horizons.
- 4 The strongest gains emerge in density forecasting, suggesting that survey information helps the model better characterise economic uncertainty.
- 5 Survey expectations not as models' competitors, rather as informative observations that can improve the estimation of forecasting models themselves.

Thank you for your attention!

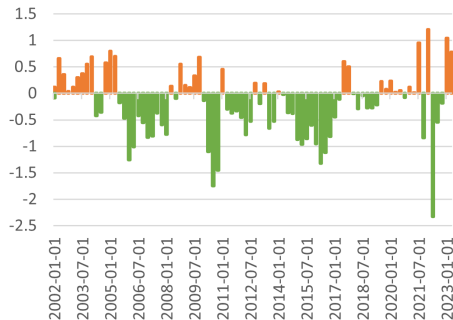
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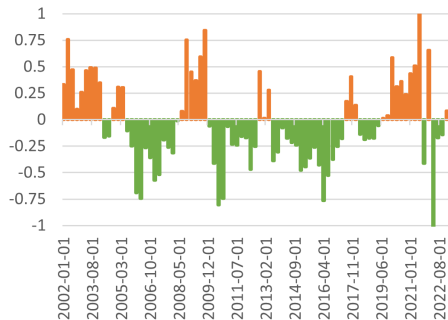
Background slides

# Forecast performance at 1-year-ahead: Real GDP



■ Model without survey is better  
■ Model with survey is better

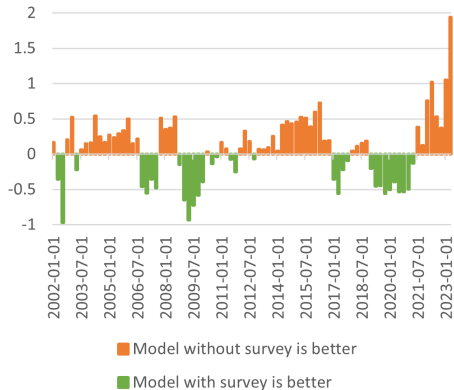
(a) Differences in RSFE



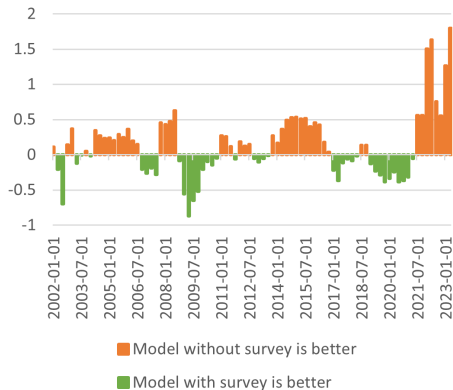
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(b) Differences in CRPS

# Forecast performance at 1-year-ahead: HICP



(a) Differences in RSFE

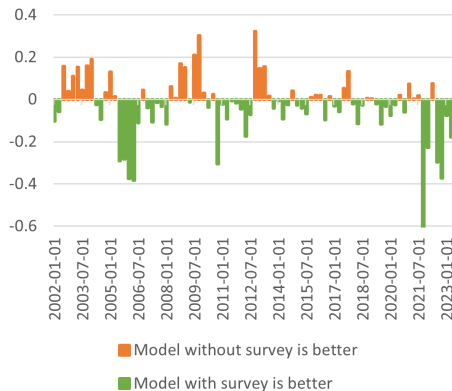


(b) Differences in CRPS

# Forecast performance at 1-year-ahead: Unemployment



(a) Differences in RSFE



(b) Differences in CRPS

$$\begin{aligned}
 & + \begin{bmatrix} 1 & 0 & \beta_{11} & \beta_{12} \\ 0 & 1 & \beta_{21} & \beta_{22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{B}_{*,2}^{[1:6,1:2]} \begin{bmatrix} y_{t+1|t} \\ \pi_{t+1|t} \\ y_{t|t} \\ \pi_{t|t} \\ y_{t-1} \\ \pi_{t-1} \end{bmatrix} = \begin{bmatrix} \mathbf{c}_{*,2} \\ \mathbf{c}_{*,1} \\ \mathbf{c}_{*,0} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{B}_{*,3}^{[1:6,1:2]} \begin{bmatrix} y_{t|t-1} \\ \pi_{t|t-1} \\ y_{t-1|t-1} \\ \pi_{t-1|t-1} \\ y_{t-2} \\ \pi_{t-2} \end{bmatrix} \\
 & + \begin{bmatrix} A_{0,11}^{-1} & A_{0,12}^{-1} & 0 & 0 & 0 & 0 \\ A_{0,21}^{-1} & A_{0,22}^{-1} & 0 & 0 & 0 & 0 \\ 0 & 0 & A_{0,11}^{-1} & A_{0,12}^{-1} & 0 & 0 \\ 0 & 0 & A_{0,21}^{-1} & A_{0,22}^{-1} & 0 & 0 \\ 0 & 0 & 0 & 0 & A_{0,11}^{-1} & A_{0,12}^{-1} \\ 0 & 0 & 0 & 0 & A_{0,21}^{-1} & A_{0,22}^{-1} \end{bmatrix} \begin{bmatrix} \exp(\lambda_{1,t+1|t}) e_{t+1|t}^y \\ \exp(\lambda_{2,t+1|t}) e_{t+1|t}^\pi \\ \exp(\lambda_{1,t|t}) e_{t|t}^y \\ \exp(\lambda_{2,t|t}) e_{t|t}^\pi \\ \exp(\lambda_{1,t-1}) e_{t-1}^y \\ \exp(\lambda_{2,t-1}) e_{t-1}^\pi \end{bmatrix}
 \end{aligned}$$

Back

$B_{*,2}$  and  $B_{*,3}$  represent the second and third power of matrix  $B_*$ , respectively:

$$B_* = \begin{bmatrix} \beta_{N \times Np} & 0_{N \times N(m-p)} \\ I_{N(m-1)} & 0_{N(m-1) \times N} \end{bmatrix} \quad (4)$$

where  $m = \max(H + 1, p)$ .

[Back](#)

For real GDP and HICP inflation, we observe:

$$X_{t,OBS}^A = \left[ \frac{\frac{1}{4} \left( Y_t^{(1)} + Y_t^{(2)} + Y_t^{(3)} + Y_t^{(4)} \right)}{\frac{1}{4} \left( Y_{t-1}^{(1)} + Y_{t-1}^{(2)} + Y_{t-1}^{(3)} + Y_{t-1}^{(4)} \right)} - 1 \right] \cdot 100 \quad (5)$$

After calculations:

$$\begin{aligned} \log \left( \frac{X_{t,OBS}^A}{100} + 1 \right) &= \frac{1}{4} \left[ \log \left( \frac{Y_t^{(4)}}{Y_t^{(3)}} \right) + 2 \log \left( \frac{Y_t^{(3)}}{Y_t^{(2)}} \right) + 3 \log \left( \frac{Y_t^{(2)}}{Y_t^{(1)}} \right) + 4 \log \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(4)}} \right) \right. \\ &\quad \left. + 3 \log \left( \frac{Y_{t-1}^{(4)}}{Y_{t-1}^{(3)}} \right) + 2 \log \left( \frac{Y_{t-1}^{(3)}}{Y_{t-1}^{(2)}} \right) + \log \left( \frac{Y_{t-1}^{(2)}}{Y_{t-1}^{(1)}} \right) \right] \end{aligned} \quad (6)$$

For real GDP and HICP inflation we observe:

$$X_{t,\text{OBS}}^Q = \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(1)}} - 1 \right) \cdot 100 \quad (7)$$

After calculations:

$$\log \left( \frac{X_{t,\text{OBS}}^Q}{100} + 1 \right) = \log \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(1)}} \right) = \log \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(1)}} \right) = \log \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(1)}} \right) = \log \left( \frac{Y_t^{(1)}}{Y_{t-1}^{(1)}} \right) \quad (8)$$



We set the law of motion of stochastic volatility to follow AR(1) process

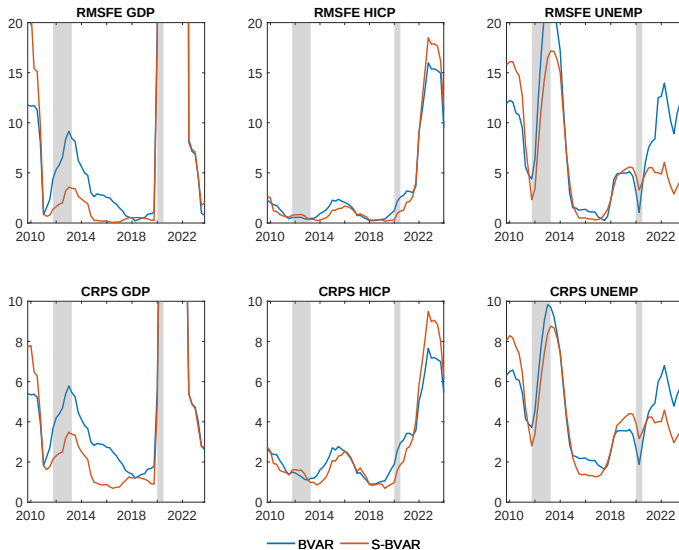
$$\lambda_t = \rho \lambda_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_u^2) \quad (9)$$

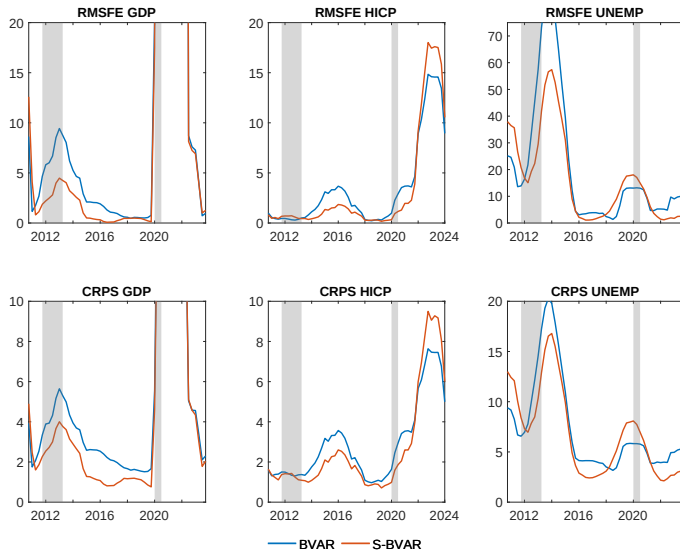
To simplify, we assume agents form expectations based on the unconditional path of volatility, that is,

$$\lambda_{t+j|t} = \rho^j \lambda_{t-1} \quad \forall j = 0, \dots, h \quad (10)$$

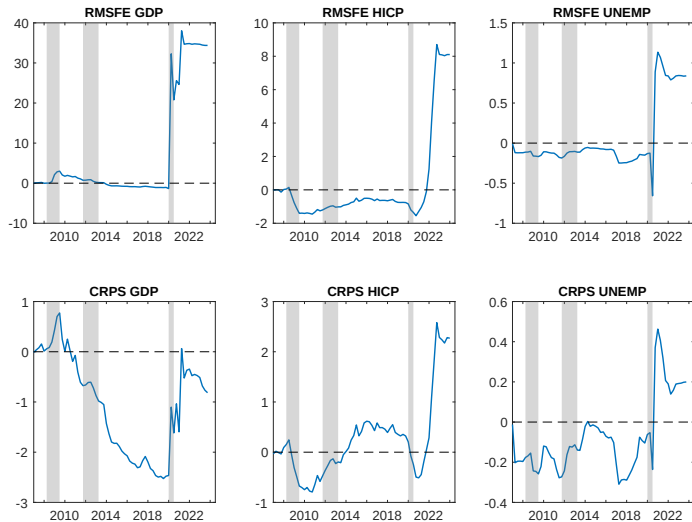
implying that respondents do not assume any conditionality about future uncertainty

$$u_{t+j|t} = 0 \quad \forall j = 0, \dots, h \quad (11)$$

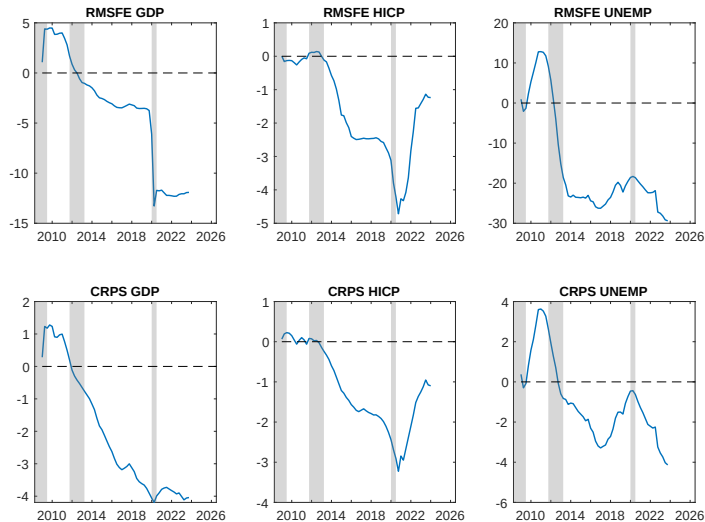




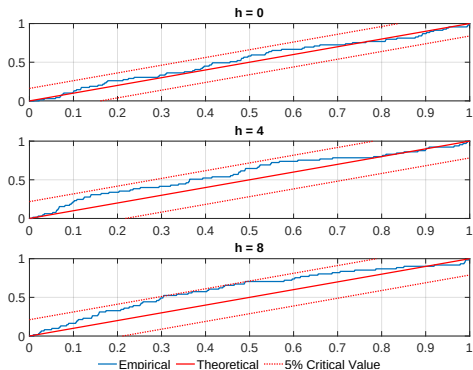
# Forecast performance at one-quarter-ahead



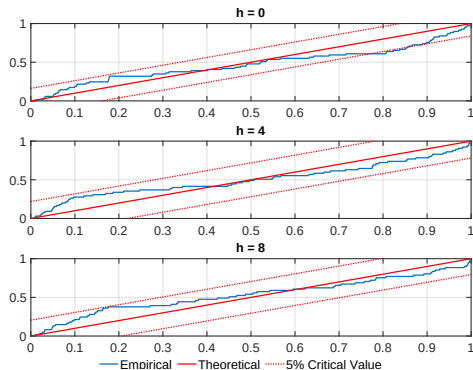
# Forecast performance at 2-year-ahead



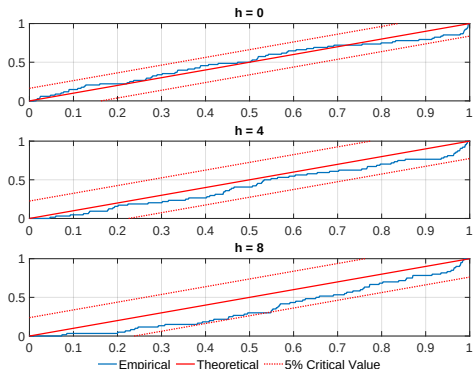
(a) Baseline



(b) S-BVAR



(a) Baseline



(b) S-BVAR

